CS341 · COMPUTER GRAPHICS

Mock Midterm Solutions

20 marks · 1 hour · same format as the past midterms

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01 Solution Q1

1. The **frame buffer** (refresh buffer, bit map) is the memory area that stores the picture of a raster system; each location corresponds to one pixel.
2. **Persistence** is the time for the emitted light to decay to one-tenth of its original intensity. **Low** persistence is better for animation.

3) $v = Q - P = (1 - 4,\ 5 - 1) = (-3,\ 4)$ $\quad |v| = \sqrt{(9 + 16)} = 5$

4) $$AR = \frac{\frac{720}{6}}{\frac{1280}{10}} = \frac{120}{128} = 0.9375$$

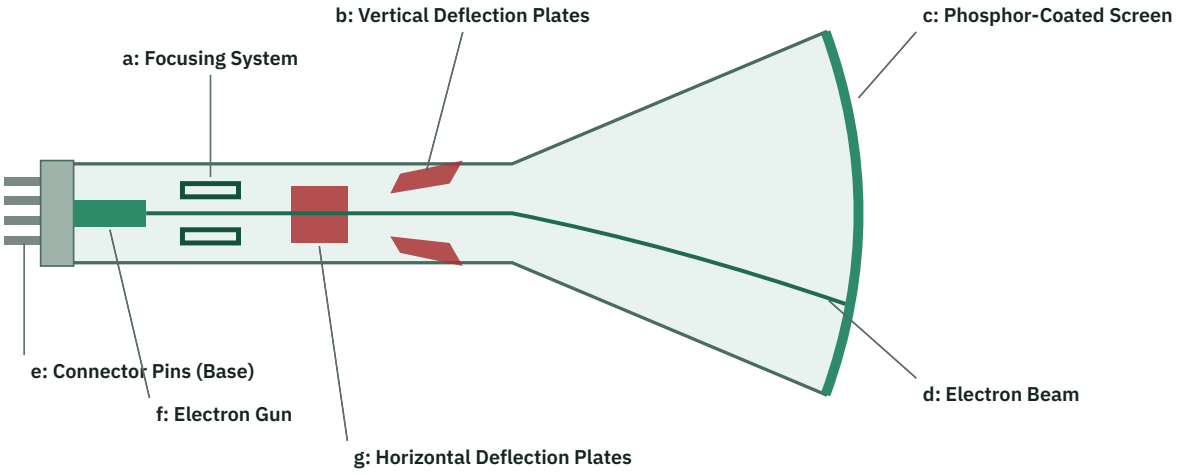
02 Solution Q2

#	Answer	Reason
1	**True**	
2	**False**	Plasma is emissive; LCD is the nonemissive one.
3	**True**	
4	**False**	The origin is black in RGB.
5	**True**	
6	**True**	Integers only.
7	**False**	That is a shade. A tint adds white.
8	**False**	The result is a scalar.

03 Solution Q3

1. b 2. c 3. b 4. c 5. b 6. b 7. b 8. b

04 Solution Q4



a: Focusing System b: Vertical Deflection Plates c: Phosphor-Coated Screen d: Electron Beam

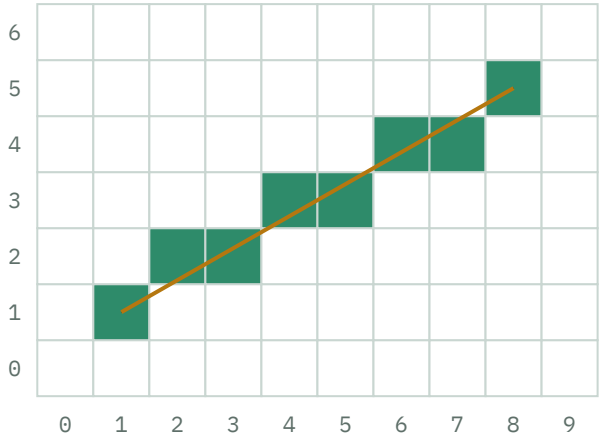
e: Connector Pins (Base) f: Electron Gun g: Horizontal Deflection Plates

05 Solution Q5

$$\Delta x = 7, \quad \Delta y = 4, \quad m = \frac{4}{7} < 1$$

$$2\Delta y = 8, \quad 2\Delta y - 2\Delta x = -6, \quad p0 = 2\Delta y - \Delta x = 1$$

k	p_k	Decision	Next pixel	p_{k+1}
0	1	$p_k \geq 0$	(2, 2)	-5
1	-5	$p_k < 0$	(3, 2)	3
2	3	$p_k \geq 0$	(4, 3)	-3
3	-3	$p_k < 0$	(5, 3)	5
4	5	$p_k \geq 0$	(6, 4)	-1
5	-1	$p_k < 0$	(7, 4)	7
6	7	$p_k \geq 0$	(8, 5)	1



Pixels: (1,1) (2,2) (3,2) (4,3) (5,3) (6,4) (7,4) (8,5)